

Task 12: Final Consultancy Recommendations for Hospital Readmission Prevention

Hospital Readmission Consultancy Team

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1 Executive Summary

This report analyses 101,766 encounter records from the Diabetes 130-US Hospitals dataset to identify readmission patterns and propose a governed Hospital Readmission Early Warning System (HREWS). The source outcome was mapped so <30 represents readmission within 30 days and >30/NO represent no readmission for binary modelling.

The analysis found an approximately 11.2% 30-day readmission prevalence. Statistical associations were observed for age, length of stay, medication count, discharge disposition, and diabetes medication status. Gender was not statistically significant in the current test. Predictive models achieved test AUC values of approximately 0.66. These results support a monitored pilot, not autonomous clinical decisions or causal claims.

2 Task 1: Industry Problem Definition

Hospital readmission affects patient safety, care continuity, staffing, and cost. The consultancy focuses on identifying patients who may benefit from coordinated discharge support while preserving clinical judgement, patient autonomy, privacy, and equity.

3 Task 2: Literature Review

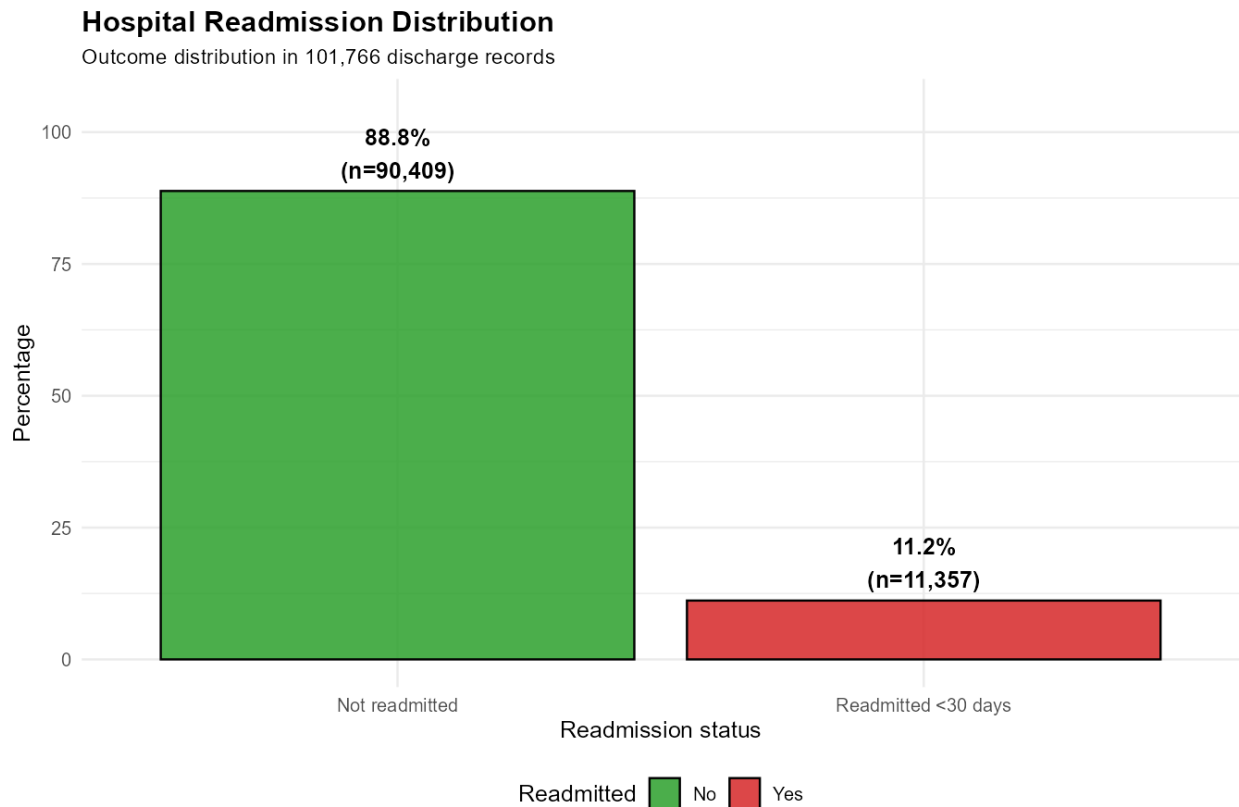
Insert the team literature review here. Include at least 15 genuine sources, consistent in citation style, with a critical synthesis rather than a list of abstracts.

4 Task 3: Descriptive Analysis

The source data contain 101,766 encounters and 50 columns. Missingness is concentrated in several administrative and clinical fields. The complete encounter cohort is retained for descriptive analysis, while each model uses the fields required for that analysis.

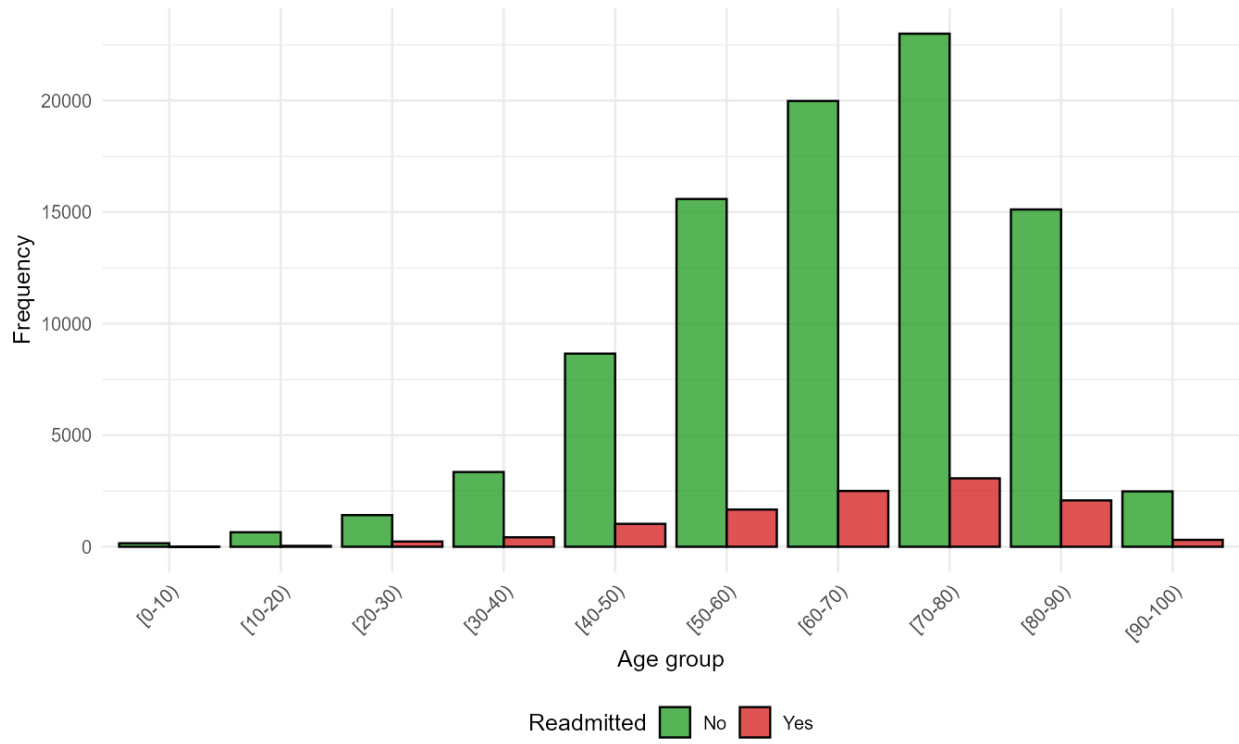
Table 1: Numeric descriptive statistics

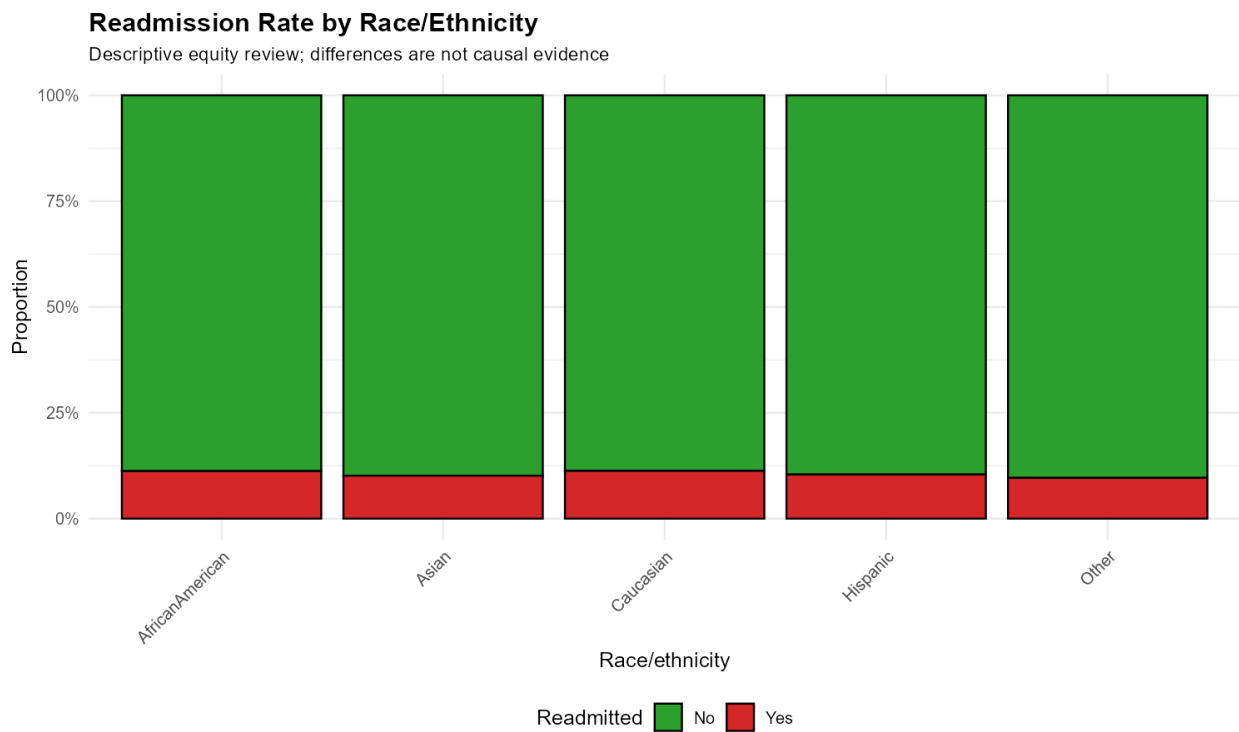
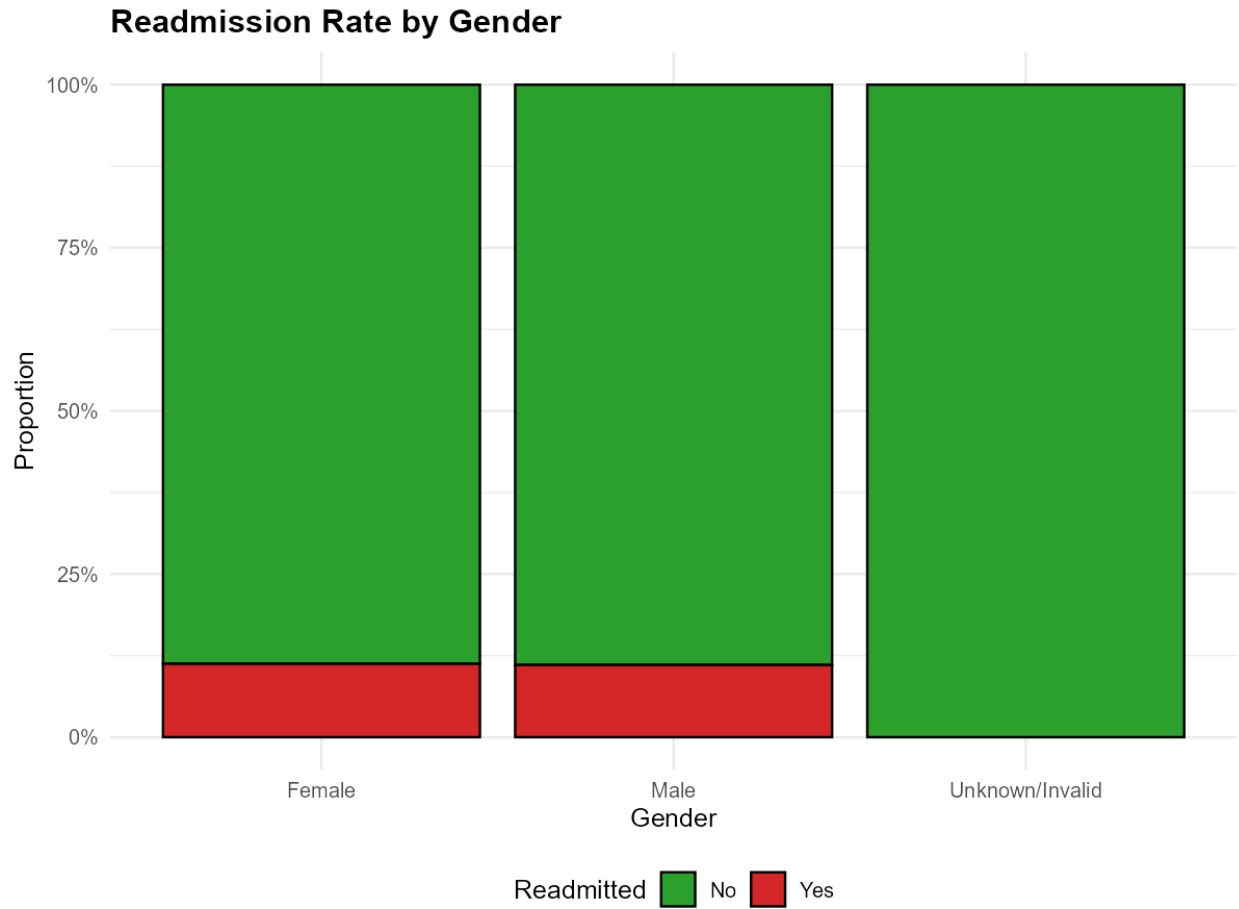
variable	n	mean	sd	median	min	max	skew	kurtosis
time_in_hospital	101766	4.40	2.99	4	1	14	1.13	0.85
num_lab_procedures	101766	43.10	19.67	44	1	132	-0.24	-0.25
num_procedures	101766	1.34	1.71	1	0	6	1.32	0.86
num_medications	101766	16.02	8.13	15	1	81	1.33	3.47
number_outpatient	101766	0.37	1.27	0	0	42	8.83	147.90
number_emergency	101766	0.20	0.93	0	0	76	22.85	1191.60
number_inpatient	101766	0.64	1.26	0	0	21	3.61	20.72
number_diagnoses	101766	7.42	1.93	8	1	16	-0.88	-0.08

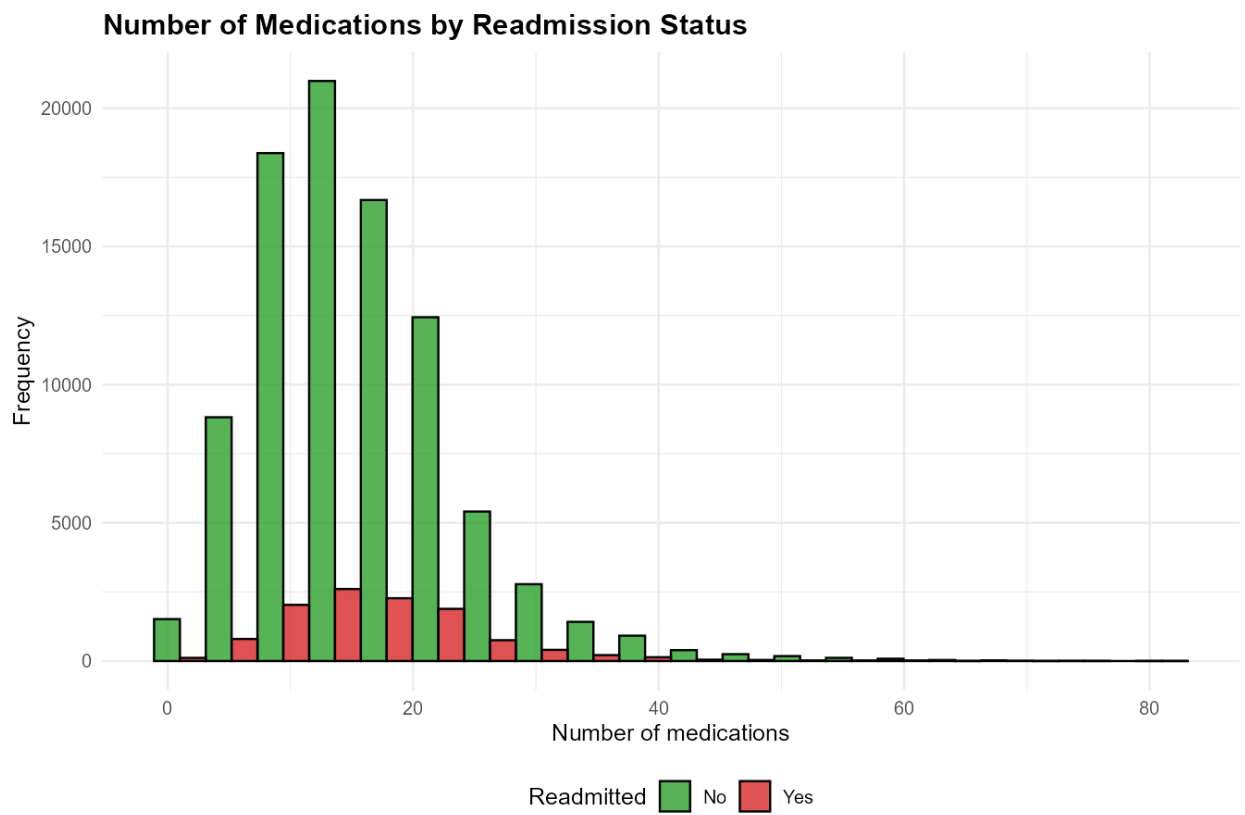
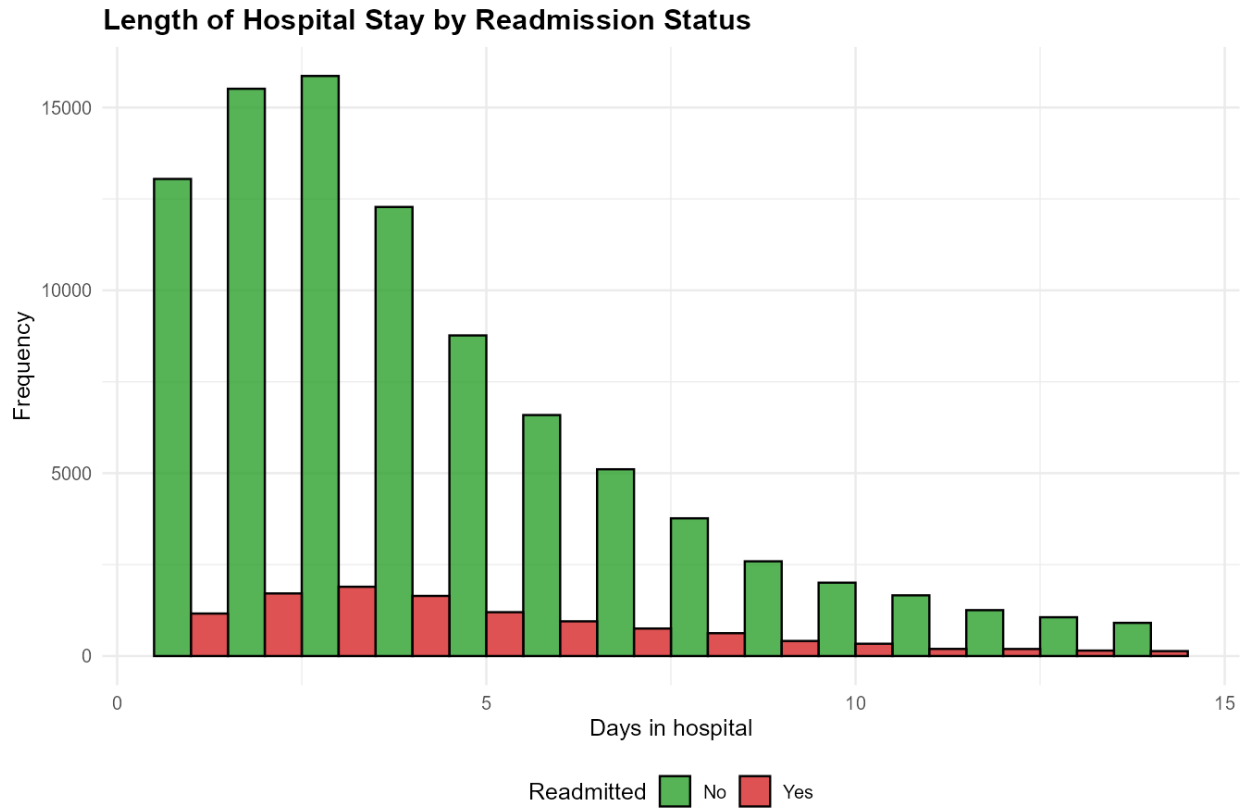


Age Distribution by Readmission Status

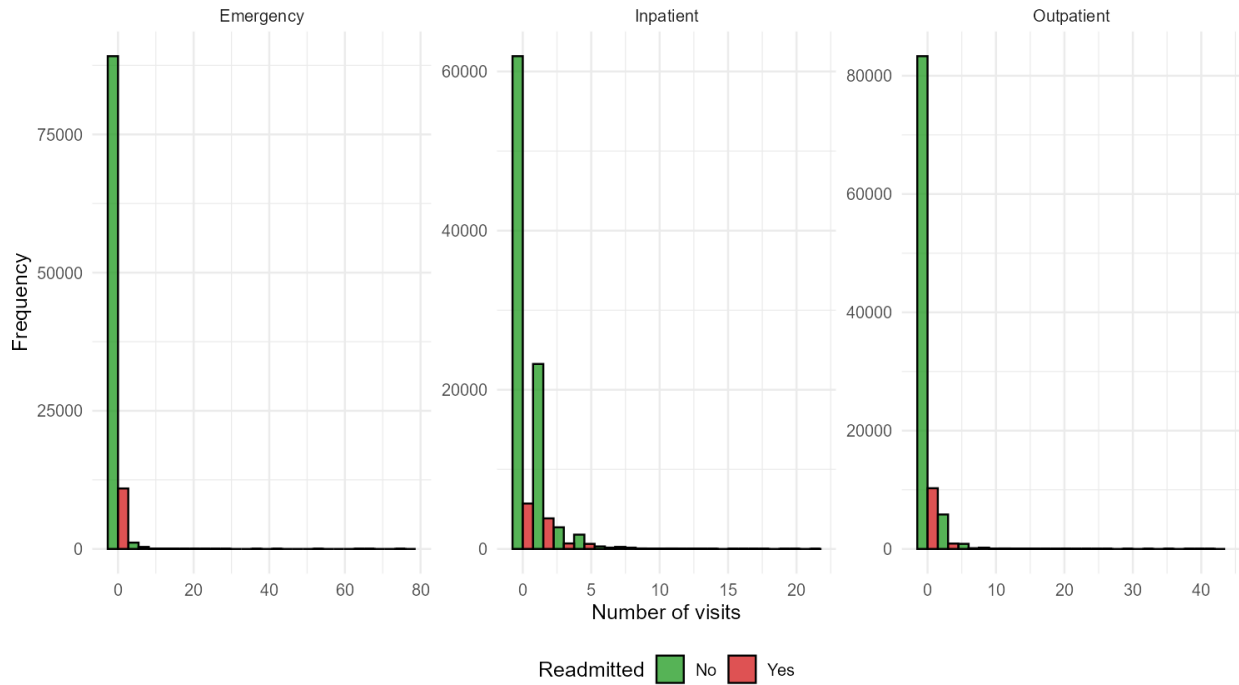
Age bands across the two outcome groups



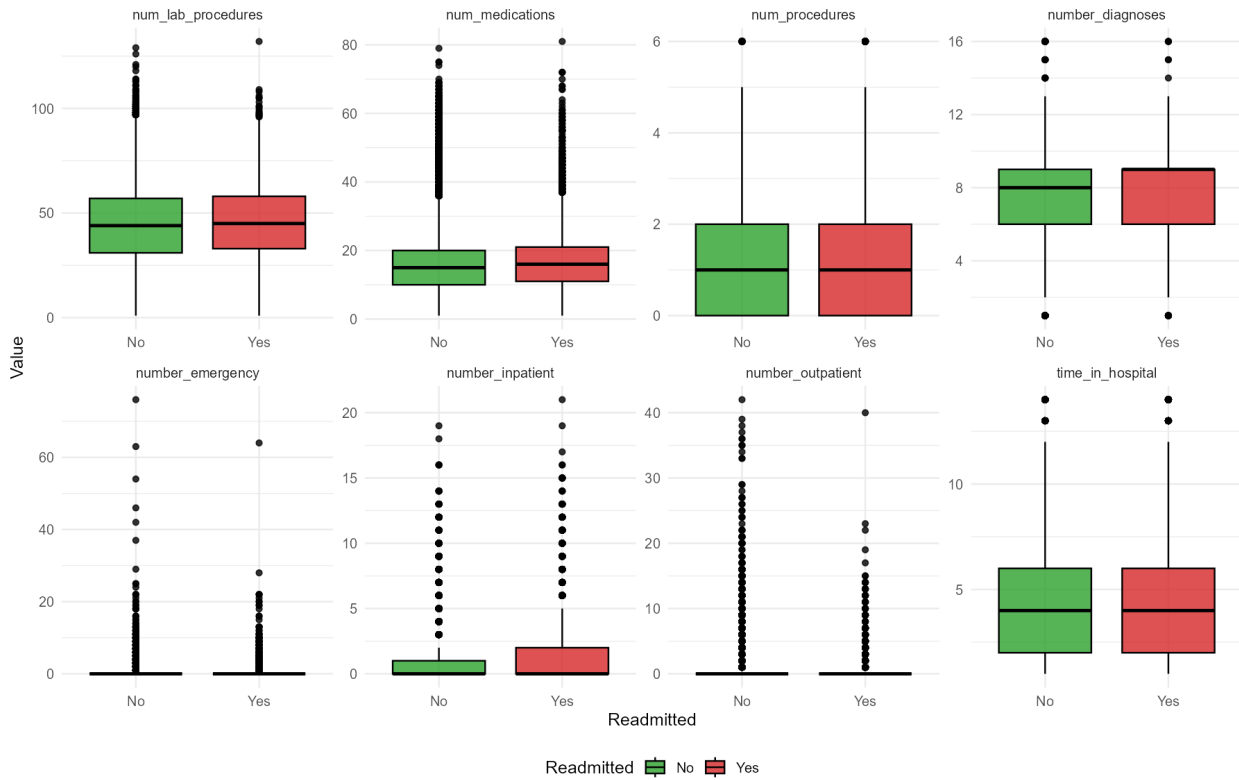




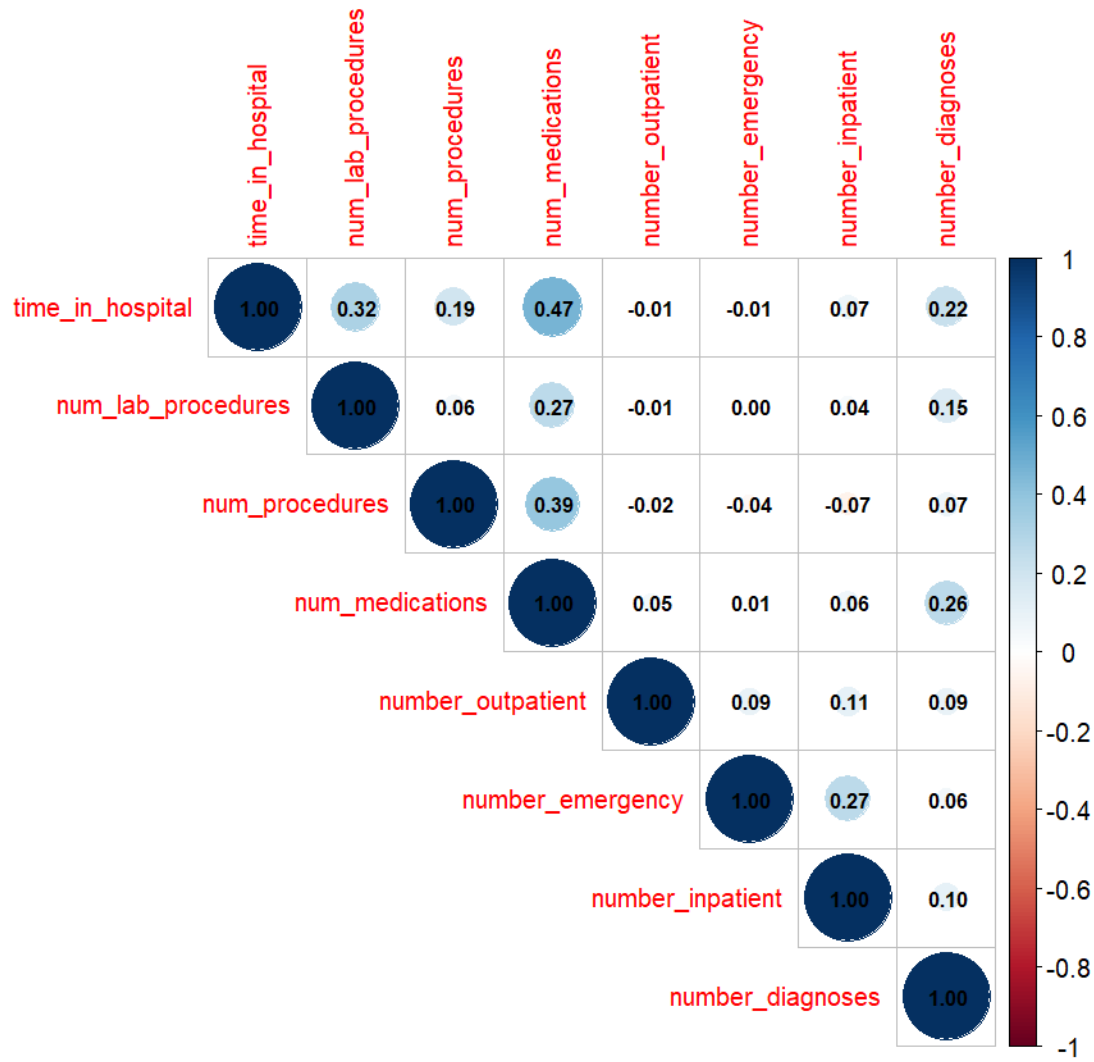
Prior Year Healthcare Utilisation by Readmission Status

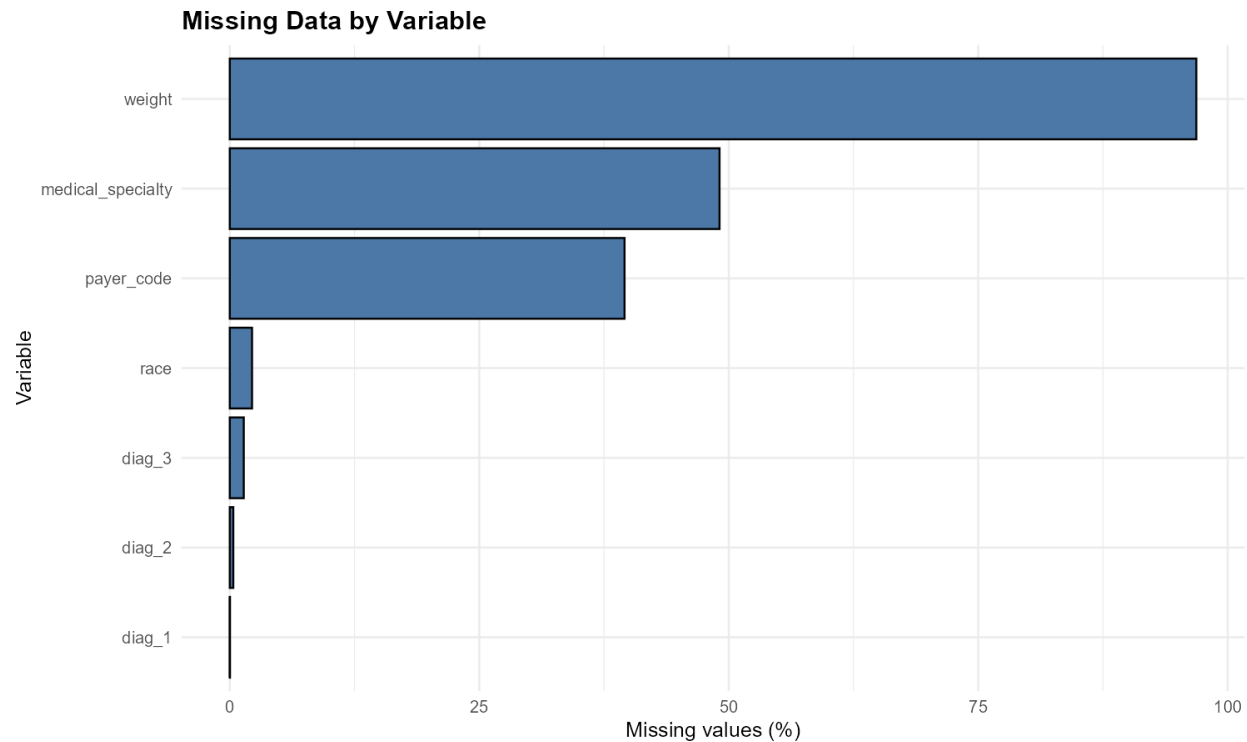


Outlier Detection: Numeric Variables by Readmission Status



Correlation Matrix: Numeric Variables





5 Task 4: Statistical Inference

The six tests are reported as associations. Welch tests were used for continuous variables, chi-square tests for categorical variables, and one-way ANOVA for discharge disposition and length of stay. Effect sizes should be considered alongside p-values because the sample is large.

Table 2: Statistical inference results

Test	Test_Type	Statistic	p-value	Significance	Effect Size	Clinical Implication
Age vs Readmission	Welch independent t-test	5.674	0.0000	Yes (p < 0.05)	0.0562	Use age-stratified discharge planning; association is not causal
Gender vs Readmission	Chi-square test of independence	1.237	0.5387	No	0.0035	Monitor subgroup performance and investigate access or care explanations
Length of Stay vs Readmission	Welch independent t-test	13.925	0.0000	Yes (p < 0.05)	0.1396	Use LOS as a complexity marker alongside clinical context

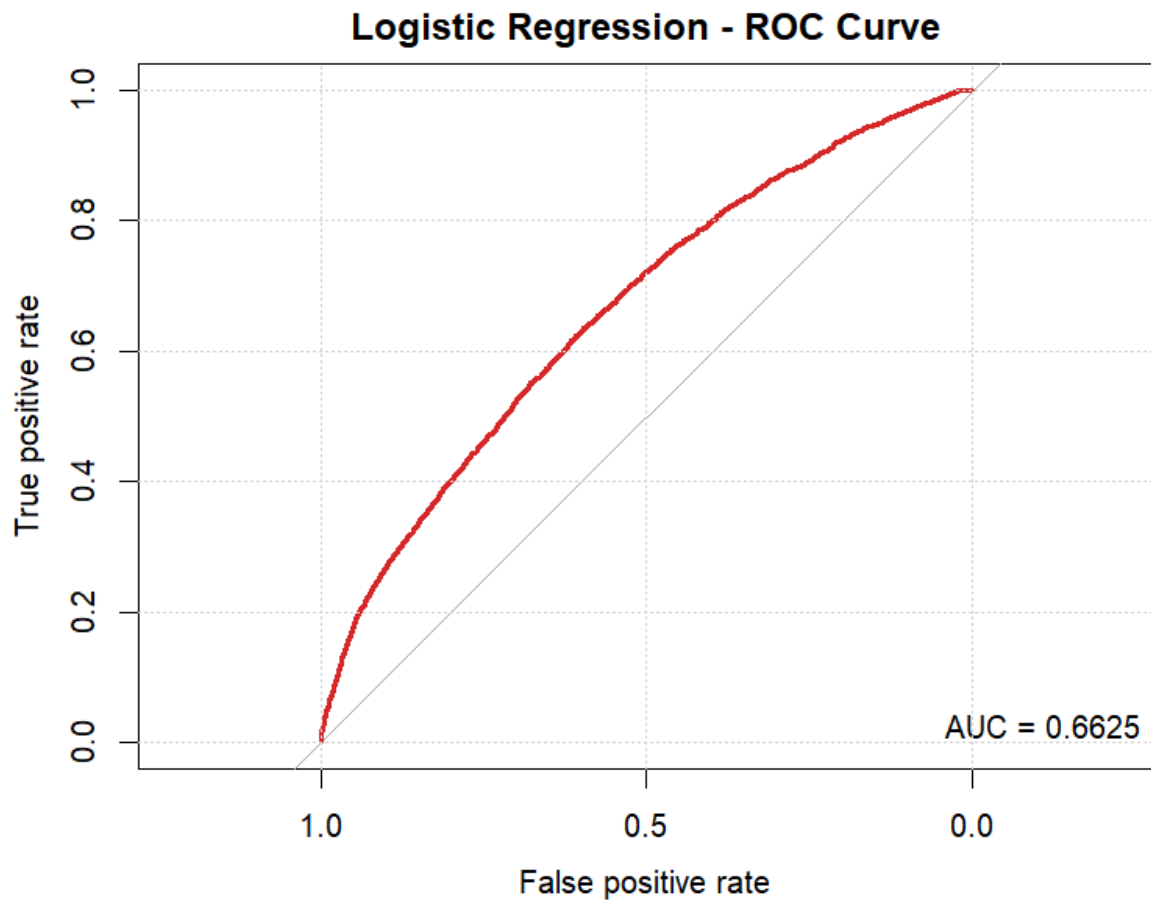
Test	Test_Type	Statistic	p-value	Significance	Effect Size	Clinical Implication
Medications vs Readmission	Welch independent t-test	12.300	0.0000	Yes (p < 0.05)	0.1223	Prioritise medication reconciliation for polypharmacy patients
Discharge Disposition vs Length of Stay	One-way ANOVA	431.98	0.0000	Yes (p < 0.05)	0.0960	Review disposition-specific discharge pathways and post-acute support
Diabetes Medication Indicator vs Readmission	Chi-square test of independence	74.66	0.0000	Yes (p < 0.05)	0.0271	Interpret medication status as a clinical complexity marker, not a treatment effect
one is the one that was used in the first one	and that was the one that was used in The first one is the one that was used in the movie The Last of Us.	NA	NA	NA	NA	

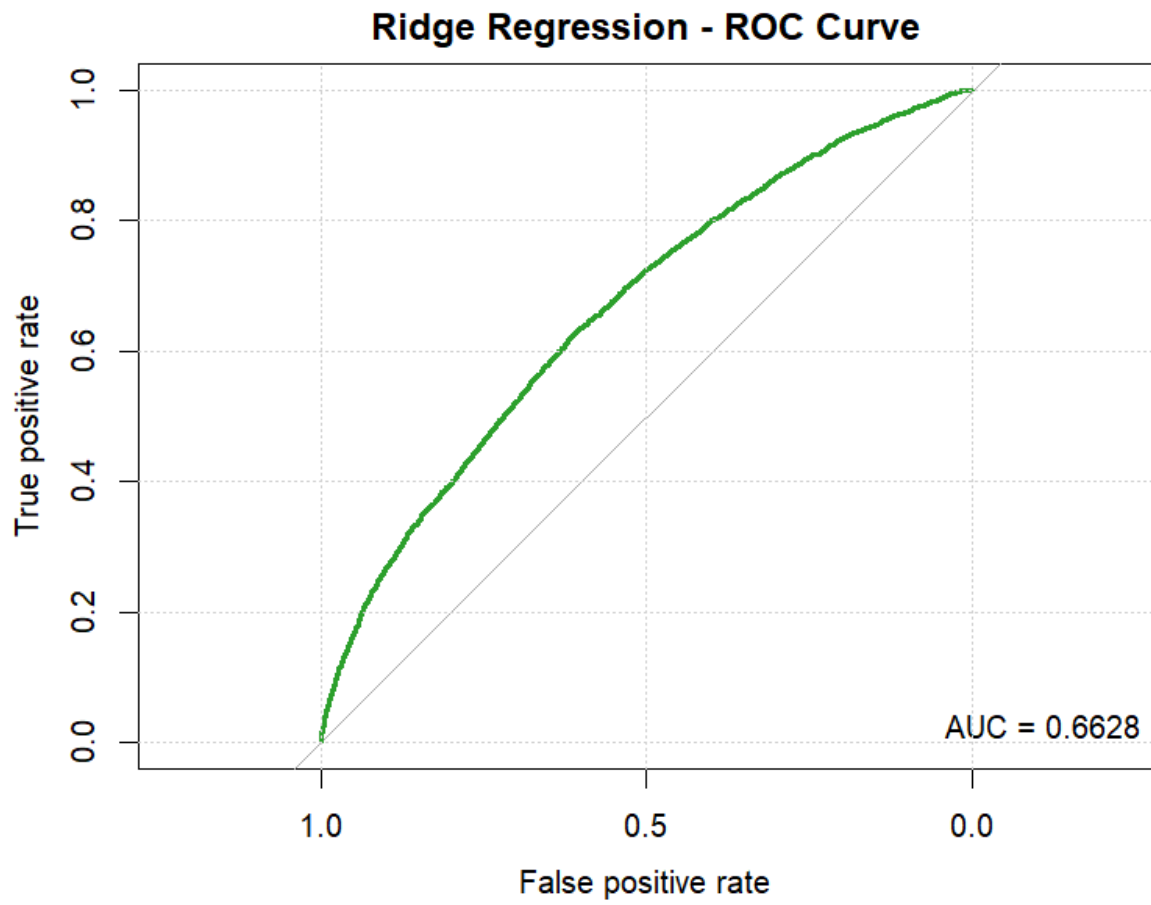
6 Task 5: Predictive Modelling

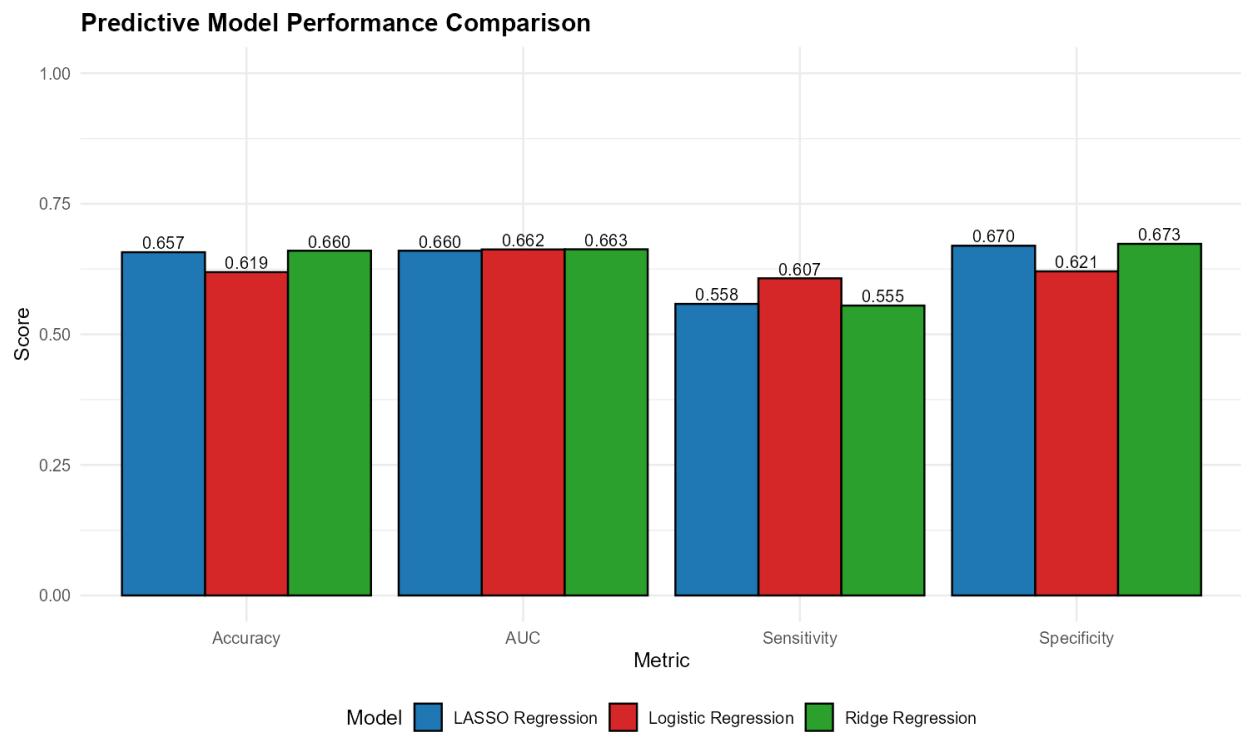
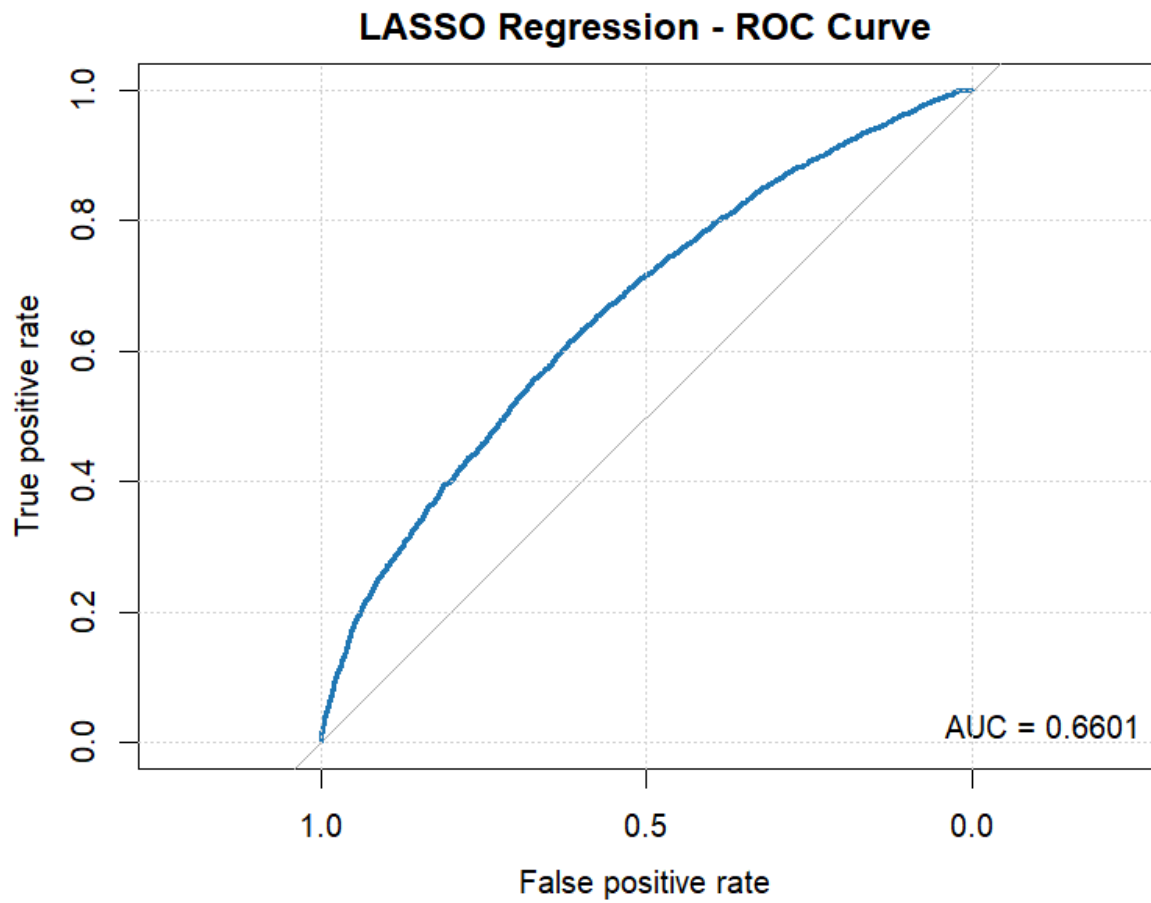
The modelling pipeline uses a stratified split, training-only standardization for penalized models, five-fold cross-validation, and thresholds selected from training ROC curves. Logistic regression provides interpretable coefficients; ridge handles correlated predictors; LASSO produces a sparse model.

Table 3: Test-set model comparison

Model	Accuracy	Sensitivity	Specificity	AUC	Threshold	Strengths	Limitations
Logistic Regression	0.6191	0.6073	0.6206	0.6625	0.1038	Interpretable coefficients	Linear log-odds assumption
Ridge Regression	0.6601	0.5551	0.6733	0.6628	0.1132	Handles correlated predictors	Retains all predictors
LASSO Regression	0.6572	0.5583	0.6697	0.6601	0.1086	Selects a sparse feature set	Selection can be unstable with correlated predictors









7 Task 6: Experimental Design

TASK 6: CRITICAL EVALUATION OF EXPERIMENTAL DESIGN Hospital Readmission Consultancy

1. CONTEXT

This consultancy uses retrospective observational discharge records. The analysis can identify associations and support risk stratification, but it cannot prove that a discharge intervention causes fewer readmissions. A future experiment should compare standard and enhanced discharge support using random assignment.

2. COMPLETELY RANDOMIZED DESIGN (CRD)

A CRD would randomly assign eligible discharged patients to two groups:

- Intervention: nurse-led follow-up call within 48 hours, medication review, and scheduled primary-care follow-up.
- Control: existing standard discharge process.
- Primary outcome: readmission within 30 days.

- Secondary outcomes: emergency visits, follow-up completion, patient experience, and intervention cost.

Advantages:

1. Randomization balances measured and unmeasured confounding on average.
2. The design is easy to communicate and analyse.
3. A single intervention can be tested with a clear endpoint.
4. The study can estimate absolute risk reduction and number needed to treat.

Limitations:

1. A single hospital or unit may limit generalisability.
2. A single intervention may not be appropriate for every risk level.
3. Patients and staff cannot usually be blinded to a follow-up call.
4. Contamination may occur if staff provide enhanced care to control patients.
5. Withholding enhanced support may raise ethical concerns; control patients must still receive standard of care.
6. RANDOMIZED COMPLETE BLOCK DESIGN (RCBD)

An RCBD would stratify patients before randomization using pre-specified risk groups, for example low, moderate, and high predicted risk. Patients would then be randomized within each block to standard or enhanced support.

Example:

- Low risk: standard discharge versus enhanced discharge checklist.
- Moderate risk: one follow-up call versus two calls plus medication reconciliation.
- High risk: standard case management versus intensive multidisciplinary follow-up.

Advantages:

1. Risk blocks reduce outcome variability and improve statistical efficiency.
2. The design tests whether intervention benefit differs by risk group.
3. It supports resource allocation because intensive care can be evaluated where expected benefit is greatest.
4. It aligns with the proposed HREWS workflow.
5. Block-specific results are directly actionable for hospital managers.

Limitations:

1. Risk cutoffs must be set before outcome analysis and validated externally.

2. Small or imbalanced blocks can reduce statistical power.
3. The analysis must include block effects and treatment-by-block interaction.
4. A risk model may reproduce historical inequities if it is not audited.
5. RCBD is more complex to operate than a simple CRD.
6. RECOMMENDED DESIGN

RCBD is preferred for a readmission-prevention pilot because the central operational question is not simply whether follow-up works. It is whether different levels of follow-up produce enough benefit for different patient-risk groups to justify their resource cost.

The trial should use intention-to-treat analysis, electronic health-record outcome capture where possible, a pre-registered analysis plan, and a sample-size calculation based on a clinically meaningful absolute reduction in 30-day readmission. Attrition and unavailable external readmissions should be considered before recruitment.

5. IMPLEMENTATION RISKS AND MITIGATION

- Loss to follow-up: use EHR, claims, telephone, and primary-care sources where permitted.
- Protocol non-adherence: use staff training, checklists, audit logs, and weekly feedback.
- Contamination: randomize at an appropriate unit or time level if patient-level allocation is impractical.
- Ethical concerns: compare standard of care with enhanced care and define stopping rules.
- Model error: validate the risk model on current hospital data before blocking.
- Equity: report recruitment, treatment receipt, and outcomes by race, gender, age, and other relevant groups.

6. CONCLUSION

CRD offers a simple causal test of one intervention. RCBD is more suitable for this consultancy because it links experimental evidence to risk-stratified clinical operations. The current observational findings should therefore be treated as a basis for designing a pilot, not as proof that HREWS will reduce readmissions.

8 Task 7: PCA Evaluation

TASK 7: CRITICAL EVALUATION OF PRINCIPAL COMPONENT ANALYSIS (PCA) Hospital Readmission Consultancy

QUESTION

Would PCA improve hospital readmission prediction for this dataset?

ASSESSMENT

PCA is not recommended as the primary modelling approach for this consultancy. The dataset has many observations but a relatively small number of clinical and utilisation predictors. The main

requirement is not merely dimension reduction; it is an interpretable and deployable risk estimate for clinical staff.

POTENTIAL BENEFITS

PCA transforms correlated numeric variables into orthogonal principal components. It could reduce the number of numeric inputs, reduce multicollinearity, and support exploratory visualization. It may be useful if future data contain hundreds of laboratory measurements, repeated observations, or high-dimensional clinical trajectories.

LIMITATIONS FOR THIS PROJECT

1. Interpretability: A component is a weighted combination of variables, not a clinically named factor. A statement such as PC1 increased risk would be less actionable than medication count or prior emergency visits increased risk.
2. Outcome relevance: PCA maximizes variance in predictors, not association with readmission. A high-variance component may have little predictive value.
3. Mixed data: Race, gender, medication indicators, and disposition are categorical. Naive PCA on numeric encodings would be inappropriate; mixed-data methods would be required.
4. Deployment: Every component score would need to be calculated and translated back into a clinical explanation.
5. Validation: Any PCA transformation must be fitted on training data only to avoid leakage.

PREFERRED ALTERNATIVE

LASSO or elastic-net regression can reduce the effective feature set while retaining original variable names. Ridge regression can handle correlated predictors without replacing them with opaque components. These approaches are better aligned with a clinical risk tool and can be compared using cross-validated discrimination, calibration, sensitivity, specificity, and decision-curve measures.

WHEN PCA WOULD BECOME REASONABLE

PCA could be evaluated in a future study with high-dimensional laboratory, imaging, or longitudinal monitoring data. It should then be compared against supervised feature selection, elastic net, and clinically interpretable baselines. The transformation must be fitted inside each resampling fold.

CONCLUSION

PCA may support exploratory analysis in a larger future data environment, but it is not the best primary method for this low-to-moderate-dimensional, clinically oriented prediction problem. Interpretable penalized regression is the preferred approach.

9 Task 8: Bayesian Methods

TASK 8: EVALUATION OF BAYESIAN STATISTICAL METHODS Hospital Readmission Consultancy

1. NAIVE BAYES

Naive Bayes applies Bayes' theorem while assuming conditional independence of predictors within each outcome class. It is fast and useful as a baseline, but the independence assumption is doubtful here because age, diagnoses, medication burden, length of stay, and prior utilisation are related. It should be used only as a comparative baseline, not assumed to be superior to logistic regression.

2. BAYESIAN LOGISTIC REGRESSION

Bayesian logistic regression assigns prior distributions to coefficients and combines them with the observed data to obtain posterior distributions. It can report a posterior mean, credible interval, and the probability that an effect is positive or negative.

Potential benefits:

- Incorporates defensible prior clinical knowledge.
- Quantifies uncertainty in coefficients and patient-level predictions.
- Supports hierarchical effects when data from multiple hospitals are available.
- Provides a natural probabilistic framework for risk thresholds and resource allocation.

Limitations:

- Prior specification requires clinical and statistical justification.
- Results should be checked under reasonable alternative priors.
- MCMC models require convergence, effective-sample-size, and posterior-predictive checks.
- A Bayesian model is not automatically better calibrated or more accurate.
- Deployment requires technical expertise and a governance process.

3. BAYESIAN DECISION MAKING

Prediction and action are different problems. A hospital can use a cost matrix to choose a threshold. If missing a patient who will be readmitted is substantially more costly than an unnecessary follow-up, a threshold below 0.50 may be reasonable. The threshold must be estimated from local costs, intervention capacity, patient preferences, and clinical safety requirements rather than asserted in advance.

Expected loss can be expressed as:

Expected loss(action) = sum over outcomes of $P(\text{outcome} \mid \text{data})$ times cost(action, outcome).

The action with the lowest expected loss is selected. This framework makes the trade-off between false negatives and false positives explicit.

4. RECOMMENDED HYBRID APPROACH

For initial implementation, retain interpretable logistic regression as the operational baseline and evaluate a Bayesian logistic model as a validation and decision-support layer. A future multi-hospital model could use hospital-specific intercepts with shared predictor effects. Priors should be weakly informative unless external evidence strongly supports a more specific prior.

Report posterior predictive discrimination, calibration, subgroup performance, credible intervals, and sensitivity to prior choices. Display uncertainty to clinicians and preserve clinical override authority.

5. CONCLUSION

Naive Bayes is suitable as a baseline. Bayesian logistic regression is valuable for uncertainty quantification, multi-hospital modelling, and cost-sensitive decisions, but it should complement rather than replace a transparent operational model until it has been validated locally.

10 Task 9: Time Series Analysis

TASK 9: TIME SERIES ANALYSIS EVALUATION Hospital Readmission Consultancy

1. CURRENT DATA LIMITATION

The available record is an encounter-level dataset with a single readmission outcome for each encounter. It does not provide a complete, time-ordered sequence of admissions, readmission dates, daily observations, or intervention dates. Traditional patient-level ARIMA or exponential-smoothing analysis is therefore not appropriate for the current table.

2. FUTURE SCENARIO A: PATIENT LONGITUDINAL DATA

If encounters could be linked over time, the project could analyse days to first readmission using survival analysis, repeated readmissions using recurrent-event or frailty models, and transitions between stable, at-risk, readmitted, and recovered states using Markov models. This would support identification of frequent users and personalized follow-up timing.

Required data include a reliable patient identifier, admission and discharge timestamps, complete readmission capture, and censoring information.

3. FUTURE SCENARIO B: HOSPITAL AGGREGATE DATA

Monthly admissions and 30-day readmission counts could form a hospital-level time series. Suitable methods include binomial or beta-binomial rate models, seasonal decomposition, exponential smoothing, ARIMA, and interrupted time-series analysis when a policy change has a known implementation date.

This is the most feasible immediate extension. It could forecast workload, identify seasonal patterns, monitor quality, and evaluate changes in discharge policy. Counts and denominators should be retained; modelling rates alone can produce misleading uncertainty.

4. FUTURE SCENARIO C: CLINICAL TRAJECTORY DATA

Daily or hourly vital signs, laboratory results, medication changes, and functional measures during an admission would permit trajectory analysis. Mixed-effects models, functional data analysis, landmark prediction, or recurrent neural networks could assess whether the direction and stability of recovery predicts safe discharge.

This requires standardized measurement, careful handling of irregular timing and missingness, and prospective validation. It should not be implemented solely because a method is technically available.

5. RECOMMENDATION

Start with monthly hospital-level aggregation if dates and denominators are available. Next, develop a prospective clinical-trajectory dataset. Patient-level recurrent-event analysis should follow only after complete longitudinal linkage is established.

6. CONCLUSION

The present data support cross-sectional risk modelling, not traditional patient-level time-series modelling. Time-series methods are promising future tools, but their validity depends on collecting the temporal structure needed by the proposed question.

11 Task 10: HREWS Innovation Proposal

TASK 10: HOSPITAL READMISSION EARLY WARNING SYSTEM (HREWS) Innovation Proposal for Data-Driven Clinical Decision Support

EXECUTIVE SUMMARY

HREWS is a proposed advisory system that estimates readmission risk at discharge, explains the main contributing factors, assigns a risk tier, and links that tier to a feasible follow-up protocol. The system is intended to support clinical judgement, not replace it.

The current analysis found modest discrimination for three interpretable penalized/logistic models, with test AUC values approximately 0.66. This is suitable for a pilot risk-stratification study, not a claim of autonomous clinical decision-making. Local validation, calibration, equity auditing, and prospective evaluation are mandatory before deployment.

1. SYSTEM COMPONENTS

A. Predictive engine

Inputs may include age band, medication burden, prior inpatient/outpatient/emergency utilisation, length of stay, procedures, diabetes medication indicator, race/gender for audit purposes, and discharge disposition. The model returns a probability and a list of model drivers.

B. Risk tiers

Initial thresholds should be selected using local intervention capacity and decision-curve analysis. Provisional thresholds can be low, moderate, and high, but they must not be treated as clinically validated cutoffs.

C. Intervention protocols

Low risk: standard discharge education, medication instructions, and routine follow-up.

Moderate risk: structured discharge checklist, medication reconciliation, primary-care communication, and a nurse call within 48-72 hours.

High risk: multidisciplinary discharge planning, intensive medication reconciliation, social-needs screening, appointment scheduling before discharge, and early nurse or telehealth contact.

D. Dashboard

Discharge planners should see risk probability, risk tier, key modifiable factors, recommended checklist, override option, and documentation fields. Managers should see calibration, readmission by tier, intervention adherence, subgroup equity metrics, and workload/cost indicators.

2. WORKFLOW

3. A discharge decision triggers an automated risk calculation.
4. The system displays probability, tier, and explanation.
5. The care team reviews the recommendation and may override it with a reason.
6. The selected intervention checklist is assigned.
7. Follow-up completion and 30-day outcomes are recorded.
8. Monthly governance reviews performance, equity, safety, and resource use.

9. EXPECTED VALUE

Potential benefits include more consistent discharge planning, earlier contact with high-risk patients, improved medication reconciliation, better coordination of post-acute services, and transparent monitoring of disparities. The size of any readmission reduction must be estimated prospectively; the observational model alone cannot establish it.

4. IMPLEMENTATION ROADMAP

Months 1-2: appoint clinical and technical governance, confirm data definitions, identify pilot unit, and obtain privacy/ethics approval.

Months 3-5: implement the baseline model, validate it on recent local data, design the dashboard, and finalize protocols.

Months 6-8: run advisory-mode pilot, measure usability, calibration, subgroup performance, intervention completion, and workflow time.

Months 9-12: expand only if pilot safety, calibration, adoption, and resource capacity targets are met.

5. GOVERNANCE AND RISK CONTROL

The system should have a named clinical owner, data owner, technical owner, and escalation process. Monitoring should include discrimination, calibration, false-negative review, subgroup performance, intervention access, data drift, and override patterns.

Risks include model drift, false reassurance, staff resistance, workflow delays, unequal intervention access, privacy breaches, and insufficient follow-up capacity. Mitigations include local validation, advisory-only deployment, audit logs, encryption, role-based access, staff training, clinical overrides, and phased implementation.

6. SUCCESS MEASURES

Primary: change in 30-day readmission rate in a prospective evaluation.

Secondary: calibration, sensitivity at an agreed operating threshold, follow-up completion, medication reconciliation completion, time to first contact, patient experience, staff workload, cost per prevented readmission, and equity of intervention receipt.

7. CONCLUSION

HREWS is a feasible innovation proposal when treated as a governed clinical decision-support pilot. The next action is not hospital-wide automation; it is local validation followed by a monitored advisory pilot and, where feasible, a randomized or controlled evaluation of the intervention protocols.

12 Task 11: Expert Validation

The interview summary is a template until a real healthcare professional has been interviewed. Replace the placeholders only with verified notes, consent status, and evidence.

EXPERT VALIDATION INTERVIEW SUMMARY

STATUS: TEMPLATE - COMPLETE ONLY AFTER A GENUINE INTERVIEW

Date: [DD/MM/YYYY] Expert Name: [Name or anonymised identifier] Title: [Job title] Organization: [Hospital/institution] Interview Format: [Phone/video/in person] Duration: [Minutes] Interviewer: [Name] Consent to document and quote: [Yes/No]

BACKGROUND

Briefly describe the consultancy, the encounter-level dataset, the main descriptive findings, the predictive models, and the proposed HREWS workflow. Do not describe an interview as completed until the conversation has occurred and evidence is retained in 06_EXPERT_VALIDATION/EVIDENCE/.

KEY FINDINGS

Q1. Problem relevance Question: What are the biggest drivers of hospital readmission in your experience? Expert response: [Record the response accurately] Our interpretation: [Link the response to the analysis]

Q2. Model validation Question: Are age, medication burden, prior utilisation, length of stay, and discharge destination plausible risk factors? What important factors are missing? Expert response: [Record the response accurately] Our interpretation: [State which factors should be emphasised or collected]

Q3. Implementation feasibility Question: Would the discharge team use an advisory risk system? What would make it useful or burdensome? Expert response: [Record the response accurately] Our interpretation: [Describe workflow implications]

Q4. Protocol appropriateness Question: Are standard, enhanced, and intensive follow-up protocols feasible for different risk levels? Expert response: [Record the response accurately] Our interpretation: [Describe required changes]

Q5. Organizational barriers Question: What are the main barriers: staffing, budget, IT integration, patient access, competing priorities, or something else? Expert response: [Record the response accurately] Our interpretation: [Connect barriers to risk mitigation]

Q6. Recommendation Question: What should the hospital do first if this system were available? Expert response: [Record the response accurately] Our interpretation: [State the prioritized next step]

OVERALL ASSESSMENT

Agreement or disagreement with the approach: [Summary] Most important expert insight: [Summary] Changes to Task 1 problem definition: [Details] Changes to Task 5 model or features: [Details] Changes to Task 10 HREWS proposal: [Details] Changes to Task 12 final recommendations: [Details]

QUOTE WITH PERMISSION

[Insert an exact quote only if permission was granted. Otherwise write: No quote used.]

EVIDENCE CHECKLIST

☐ Interview notes or transcript retained ☐ Consent documented ☐ Evidence file added to 06_EXPERT_VALIDATION/EVIDENCE/ ☐ Names and patient-identifying information handled appropriately ☐ Claims in the report reflect the actual interview

13 Task 12: Final Recommendations

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{r setup, include=FALSE} knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)
```

14 Executive Summary

The analysis used 101,766 encounter records from the Diabetes 130-US Hospitals dataset. The target was mapped from the source values: <30 became readmitted within 30 days, while >30 and NO were treated as not readmitted for the binary modelling task. Approximately 11.2% of encounters were readmitted within 30 days, creating a substantial class imbalance.

Descriptive analysis identified differences in age, length of stay, medication count, discharge disposition, and diabetes medication status. Gender was not statistically significant in the current analysis. These findings are associations in retrospective data and should not be interpreted as causal effects.

The corrected modelling pipeline used a stratified split, training-only standardization for penalized models, five-fold cross-validation, and thresholds selected from training data. Test AUC was approximately 0.66 for logistic, ridge, and LASSO models. The models are suitable for a monitored pilot, not autonomous clinical decisions.

15 Recommendations

15.1 Recommendation 1: Validate an advisory HREWS pilot

Implement a Hospital Readmission Early Warning System in advisory mode for one clinical unit. Show a probability, provisional risk tier, key drivers, and a recommended checklist. Require clinician review and permit documented overrides.

Before deployment, validate discrimination and calibration on recent local data. Monitor false negatives, data drift, subgroup performance, intervention access, and workflow time.

15.2 Recommendation 2: Match follow-up intensity to need

Use three operational tiers only after local thresholds are validated.

- Low: standard education, medication instructions, and routine follow-up.
- Moderate: structured checklist, medication reconciliation, primary-care communication, and a 48-72 hour call.
- High: multidisciplinary planning, social-needs screening, early nurse or telehealth contact, and scheduled specialist follow-up.

The intervention must be resourced. A model cannot reduce readmissions if the organization cannot complete the recommended follow-up.

15.3 Recommendation 3: Establish equity and safety governance

Report calibration, sensitivity, specificity, precision, and intervention completion by relevant demographic groups. Review whether the system underpredicts risk or allocates fewer services to any group. Use race and gender primarily for auditing unless there is a defensible clinical reason to use them for prediction.

Create a monthly review group with clinical, analytics, IT, quality, finance, and patient representation. Clinical judgement remains authoritative.

15.4 Recommendation 4: Test causal effectiveness

The current work is observational. A prospective randomized complete block design, with randomization within validated risk strata, can test whether enhanced follow-up causes a reduction in 30-day readmission. Use intention-to-treat analysis, a pre-specified outcome, local sample-size calculations, and standard-of-care controls.

15.5 Recommendation 5: Build better temporal data

The present table is not a patient-level time series. Future work should first aggregate monthly admissions and readmission counts if valid dates are available. Later work can link longitudinal admissions and collect daily clinical trajectories for survival, recurrent-event, or trajectory modelling.

16 Implementation Timeline

Months 1-2: governance, data definitions, local validation, privacy review, and pilot selection.

Months 3-5: dashboard prototype, workflow design, staff training, intervention protocols, and monitoring plan.

Months 6-8: advisory pilot with usability, calibration, equity, and workload evaluation.

Months 9-12: controlled expansion only if safety, capacity, and outcome criteria are met.

17 Limitations

The dataset is retrospective and encounter-level. It lacks many social determinants, complete longitudinal history, intervention exposure, and reliable temporal structure for patient-level time series analysis. Missingness is concentrated in variables such as weight, payer code, and specialty. Predictive performance may not transfer to another hospital. Statistical significance is not equivalent to clinical importance, and no observed association establishes that an intervention will reduce readmission.

18 Conclusion

The most defensible action is a governed, locally validated, advisory HREWS pilot linked to risk-matched follow-up and equity monitoring. Prospective controlled evaluation should determine whether the proposed interventions improve outcomes before broad deployment.

19 References

Insert the genuine 15-plus literature sources here in Harvard or APA format. Check every statistic and external claim against its source before submission.